# Cross-Modal Attention Fusion for Anterior Segment Segmentation Using Complementary Image Representations


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***Abstract*—Accurate segmentation of anterior segment structures — including the cornea, pupil, and lens — is clinically relevant for cataract assessment and surgical planning. Unimodal approaches that operate solely on RGB images are sensitive to lighting artifacts and the low color contrast caused by lens opacification, limiting their reliability in real-world ophthalmic settings. We propose a dual-encoder architecture that fuses complementary representations of the same image to address this limitation. One encoder processes the original RGB image to capture photometric and textural features, while a parallel encoder processes a structural edge map derived via Canny filtering to capture geometric boundary information. A cross-attention module at the bottleneck and at encoder stage 4 allows the RGB encoder to query the edge encoder's feature space, enriching the primary representation with explicit structural cues before decoding. Although both inputs originate from the same image, they occupy heterogeneous feature spaces — photometric and geometric — qualifying the approach as multimodal fusion in the architectural sense. We evaluate the proposed model on the Cataract-Seg dataset against unimodal U-Net baselines and an early-fusion concatenation baseline, reporting Intersection over Union (IoU) and Dice coefficient as primary metrics. The proposed cross-attention fusion strategy achieves competitive performance with strong RGB baselines, matching the best result (Early Fusion, IoU 0.9532) at 0.9530 IoU, while providing a more principled and interpretable fusion mechanism. The near-ceiling performance across RGB-based models indicates dataset saturation; the gated residual design ensures the model does not degrade when edge maps are noisy, making it a practical choice for resource-constrained clinical deployments relative to large foundation model adaptations.**

***Index Terms*—cataract segmentation, cross-modal attention, anterior segment, deep learning, multimodal fusion**


## I. Introduction

Cataract disease affects over 100 million people worldwide and remains the leading cause of preventable blindness [1]. Accurately segmenting anterior segment structures , the cornea, pupil, and lens , from clinical images is essential for automated grading, surgical planning, and intraoperative guidance. Segmentation refers to assigning a class label to every pixel in the image, producing a binary mask that indicates which pixels belong to the anatomical structures of interest. This pixel-level classification is a fundamental step in computer-aided diagnosis systems for ophthalmic imaging. However, models that process only the RGB image struggle with two problems: the low color contrast caused by lens opacification (the lens becomes cloudy, making its boundary hard to see) and the illumination artifacts common in slit-lamp photography. These issues reduce reliability precisely in the cases that matter most clinically.

A natural solution is to give the model a second, complementary view of the same image. A *Canny edge map* provides exactly that: it detects boundaries by locating pixels where brightness changes sharply. These boundaries correspond to anatomical edges (cornea boundary, lens capsule) that remain visible even when color contrast is lost. The RGB image carries color and texture information, while the edge map carries boundary information. Although both come from a single photograph, they represent fundamentally different types of information, which makes their fusion a *multimodal* problem in the architectural sense.

Multimodal fusion architectures that combine complementary representations have proven effective in other medical domains. In brain tumor segmentation, cross-attention between different MRI modalities (T1, T2, FLAIR) consistently outperforms both single-modality and early-concatenation baselines [2]. Dual-encoder designs pairing convolutional networks with transformers have shown similar gains [3]. However, these methods assume that multiple modalities come from separate sensors. In ophthalmic imaging, obtaining additional modalities (e.g., OCT alongside a slit-lamp photograph) is not always feasible, and no prior work has explored creating a second representation from a single cataract image.

We propose a dual-encoder U-Net that processes the RGB image through one encoder and its Canny edge map through a second, independent encoder. Cross-attention modules at two stages let the RGB features "query" the edge features for boundary information, enriching the RGB representation with explicit structural cues. A learnable scalar gate controls how much edge information flows in , it starts turned off (initialized so that $\sigma(\theta) \approx 0.02$) so the model initially behaves like a standard RGB U-Net, and gradually learns to rely on edge cues only where they help. Edge encoder skip connections further propagate boundary information through every level of the decoder.

We evaluate on the Cataract-Seg dataset [4] against three baselines: a standard RGB U-Net, an edge-only U-Net, and an early-fusion U-Net that concatenates both inputs at the first layer. The proposed model achieves an IoU of 0.9530, matching the strongest baseline, while providing a more principled and interpretable fusion mechanism than simple input concatenation.

## II. Related Work

### A. Anterior Segment and Cataract Segmentation

Encoder-decoder architectures have become the standard approach for medical image segmentation since Ronneberger et al. introduced U-Net [5]. U-Net uses a downsampling path followed by a symmetric upsampling path connected by skip connections, which enables precise localization even with very few annotated images. The skip connections transfer fine spatial detail from the encoder directly to the corresponding decoder stages, addressing the loss of spatial resolution that occurs during downsampling. This design has been widely adopted in ophthalmic segmentation because it preserves the fine anatomical boundaries needed for accurate delineation of corneal and lens structures.

Surgical scene understanding in cataract procedures presents additional challenges due to reflections, occlusions from instruments, and changes in illumination during surgery. The CaDIS dataset [6] addressed this gap by providing annotated frames of cataract surgery videos covering anatomical structures and surgical instruments, establishing a benchmark for multi-class segmentation in this domain.

### B. Multimodal Fusion in Medical Imaging

The transformer architecture introduced by Vaswani et al. [7] established attention mechanisms as a way to model relationships between different parts of a sequence. Its key insight , that queries from one representation can attend to keys and values from another , directly motivates the cross-modal fusion strategy we adopt at the bottleneck.

In the medical imaging domain, Zhang et al. proposed mmFormer [2], a multimodal medical transformer for brain tumor segmentation that can handle cases where some MRI modalities are missing. mmFormer uses separate encoders for each modality and connects them with attention mechanisms, showing that cross-modal attention outperforms simply concatenating all modalities at the input , especially when different modalities capture very different types of information.

Li et al. proposed DECTNet [3], a dual-encoder architecture combining a convolutional branch with a Swin Transformer branch, fusing their complementary representations via a feature fusion decoder. This confirms that running two parallel encoders and merging their outputs is an effective strategy for medical segmentation. In contrast to DECTNet, which uses two different network architectures processing the same input, our approach uses two identical encoders processing structurally different representations (color and geometry) of the same image.

The multi-modal ophthalmic AI survey by Wang et al. [1] reviewed deep learning across fundus photography, OCT, OCTA, and clinical metadata, identifying cross-modal feature fusion as an open research direction with clinical impact. This motivates our exploration of multimodal fusion for cataract segmentation specifically.

### C. Large Pre-Trained Models for Ophthalmic Segmentation

Recent advances in large pre-trained models have changed medical image segmentation. Ma et al. introduced MedSAM [8], fine-tuning the Segment Anything Model (SAM) on over 1.5 million medical image-mask pairs across 10 imaging modalities. While MedSAM achieves strong generalization, its scale , requiring 20 A100 GPUs for training , makes it impractical for deployment on small ophthalmic datasets without substantial computational infrastructure.

The challenge of adapting large models to specific tasks was addressed by Chen et al. with SAM-Adapter [9], which introduces lightweight task-specific adapters to guide SAM toward domain-specific features. While adapter-based methods reduce fine-tuning cost, they still inherit the full SAM encoder and are optimized for scenarios with large pre-training data available.

### D. Positioning of the Proposed Work

The above literature reveals three complementary trends: (1) task-specific architectures like U-Net achieve strong segmentation performance but only have access to the information in a single image modality; (2) cross-modal attention mechanisms can effectively fuse different types of information but have not been applied to combining an image with its own edge map for cataract segmentation; and (3) large pre-trained models provide broad generalization but require computational resources that are impractical for small, specialized datasets.

Our proposed architecture bridges the gap between (1) and (2): we adapt the dual-encoder, cross-attention paradigm demonstrated in mmFormer and DECTNet to a lightweight model where the two encoders process the original RGB image and its Canny edge map respectively. This design explicitly provides the decoder with complementary color and geometric features through a bottleneck cross-attention module, without requiring large-scale pre-training. Compared to large pre-trained model adaptation (3), this approach is computationally accessible and directly optimized for the structural characteristics of cataract anterior segment images.

## III. Proposed Method

### A. Overview

We propose a dual-encoder segmentation architecture that fuses two complementary representations of the same input image: the original RGB image and a structural edge map derived from it via Canny filtering. Although both inputs originate from a single image, they occupy different feature spaces , color and texture versus geometry , and are processed by independent encoders whose representations are fused through cross-attention modules at multiple scales.

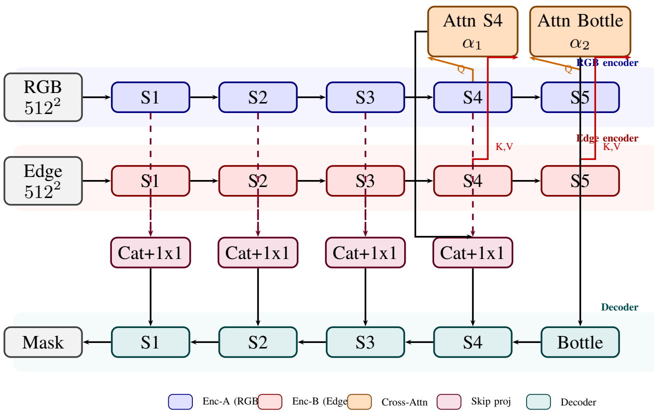


Fig. 1. Proposed dual-encoder architecture. Blue: RGB encoder (Enc-A). Red: Edge encoder (Enc-B). Orange: Cross-attention fusion modules with learnable gates $\alpha_1$, $\alpha_2$. Purple: fused skip projections ($1\times1$ conv). Green: decoder. Dashed arrows indicate edge skip connections concatenated with RGB skips before projection.

### B. Input Representations

Given an input image $\mathbf{I} \in \mathbb{R}^{H\times W\times 3}$, two representations are constructed:

**RGB stream.** The original image is normalized to $[0, 1]$ and resized to $512 \times 512$.

**Edge stream.** A single-channel edge map $\mathbf{E} \in \mathbb{R}^{H\times W\times 1}$ is computed by applying Canny edge detection to the grayscale conversion of $\mathbf{I}$. The Canny algorithm works in four steps: (1) smooth the image with a Gaussian filter to remove noise, (2) find the intensity gradient at every pixel (how fast and in what direction brightness changes), (3) thin the resulting edges by keeping only pixels that are local maxima in the gradient direction, and (4) link edge pixels into continuous contours using two thresholds , pixels above $t_2 = 150$ are kept as strong edges, pixels below $t_1 = 50$ are discarded, and pixels in between are kept only if they connect to a strong edge. This dual-threshold strategy produces clean, connected boundaries while suppressing noise. After detection, the edge map is replicated to three channels to match the expected input shape of the encoder. This representation makes the geometric boundary structure explicit, which is clinically relevant for lens delineation under variable illumination.

### C. Dual-Encoder Architecture

Both streams are processed by independent U-Net encoders sharing the same ResNet-34 architecture but no weights. ResNet-34 consists of four residual blocks with channel depths 64, 64, 128, and 256, followed by an average pooling layer that produces 512-channel features. Each residual block uses bottleneck layers with batch normalization and ReLU activations, and skip connections that add the block input to its output, enabling training of deeper networks without vanishing gradients. The encoder follows the standard downsampling path of five resolution stages, producing feature maps at channel depths 64, 64, 128, 256, and 512 respectively. For a $512\times512$ input, the bottleneck feature map has spatial resolution $16\times16$.

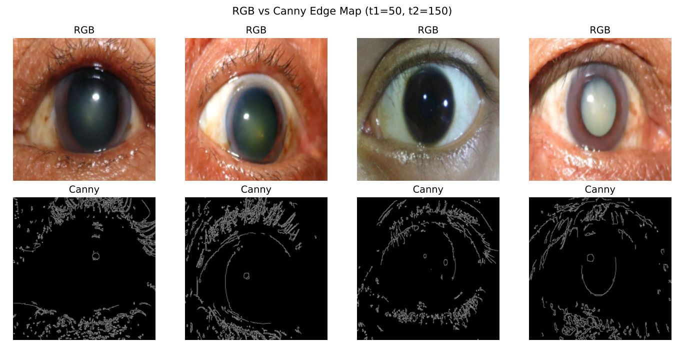


Fig. 2. Sample anterior segment images (top) and their corresponding Canny edge maps (bottom, $t_1$=50, $t_2$=150). The edge stream makes lens and corneal boundaries geometrically explicit even when color contrast is low.

Let $\mathbf{F}_A^{(s)}$ and $\mathbf{F}_B^{(s)}$ denote the feature maps from the RGB and edge encoders at stage $s$, with the bottleneck at stage 5 ($C = 512$) and stage 4 at ($C = 256$, spatial $32\times32$).

### D. Cross-Attention Fusion Module

The bottleneck features are fused via multi-head cross-attention [7]. Both feature maps are reshaped from $(B, C, H', W')$ to sequence format $(B, H'W', C)$. The RGB features serve as queries and the edge features supply keys and values:

$$\tilde{\mathbf{F}}_A = \text{LayerNorm}(\alpha \cdot \text{MHA}(Q{=}\mathbf{F}_A,\ K{=}\mathbf{F}_B,\ V{=}\mathbf{F}_B) + \mathbf{F}_A) \tag{1}$$

Intuitively, this attention operation lets each spatial location in the RGB feature map “look at” all spatial locations in the edge feature map and decide which boundary cues are relevant at that position. The queries $\mathbf{Q}$ from the RGB features represent “what boundary information do I need here?” while the keys $\mathbf{K}$ from the edge features represent “what boundaries are present at each location?” The dot product $\mathbf{Q}\mathbf{K}^\top$ produces an attention map that determines how much each edge location should influence each RGB location.

The multi-head mechanism applies this attention computation in parallel across $h = 8$ independent heads. Each head learns a different projection of the queries, keys, and values, allowing the model to attend to different types of cross-modal relationships simultaneously. The outputs of all heads are concatenated and projected back to the original feature dimension, producing the final attention output. This multi-head design has been shown to improve the expressiveness of attention mechanisms by capturing diverse relational patterns.

The attention output is then added to the original RGB features through a residual connection, followed by layer normalization. The residual connection ensures that the RGB features are not overwritten but rather enriched by the edge information, preserving the original photometric representation while incorporating geometric cues.

where $\alpha = \sigma(\theta)$ is a learnable scalar gate with $\theta$ initialized to $-4$, giving $\alpha \approx 0.02$ at the start of training. This allows the model to begin as a near-pure RGB U-Net and gradually learn to incorporate edge information only when it is beneficial , an important safeguard given the noisy nature of Canny edge maps. Intuitively, the gate acts as a volume knob for edge information: it starts at zero so the model is not forced to use potentially noisy edge features, and training can turn it up only if the edge cues consistently improve the segmentation.

### E. Multi-Scale Fusion

Fusing edge information only at the bottleneck (the lowest, most compressed resolution) means the decoder receives boundary cues only after they have been heavily downsampled. Fine-grained edge details lost during downsampling cannot be recovered. We address this with two additional mechanisms that deliver edge information at multiple resolutions:

**Stage-4 cross-attention.** A second cross-attention module (256 channels, $n_{\text{heads}} = 8$, with its own learnable gate) is applied one encoder stage above the bottleneck (32×32 spatial resolution). This injects geometric cues into the feature hierarchy before the bottleneck, providing richer structural context at an earlier stage of the downsampling path.

**Edge encoder skip connections.** At each of the four skip-connection levels in the decoder, the corresponding edge encoder feature map $\mathbf{F}_B^{(s)}$ is concatenated with the RGB skip $\mathbf{F}_A^{(s)}$ along the channel dimension, then projected back to the original channel count via a 1×1 convolution:

$$\hat{\mathbf{F}}^{(s)} = \text{Conv}_{1\times1}\left(\left[\mathbf{F}_A^{(s)} \parallel \mathbf{F}_B^{(s)}\right]\right), \quad s \in \{1, 2, 3, 4\} \tag{2}$$

This propagates geometric boundary cues through the full decoder path rather than only at the bottleneck, enabling boundary-aware feature refinement at every decoder resolution.

### F. Decoder and Output

The decoder follows the standard U-Net expanding path with bilinear upsampling and two 3×3 convolutional blocks at each stage. At each decoder level, the upsampled feature map from the previous stage is concatenated with the corresponding fused skip connection $\hat{\mathbf{F}}^{(s)}$, then passed through two convolutions with batch normalization and ReLU activation. This design follows the original U-Net formulation and ensures that fine spatial details from the encoder are combined with semantic information from the decoder at every resolution.

The fused skip connections $\hat{\mathbf{F}}^{(s)}$ replace the plain RGB skips at each resolution level. A final 1×1 convolution with sigmoid activation produces the binary segmentation mask $\hat{\mathbf{M}} \in [0, 1]^{H\times W}$.

### G. Training Objective

The model is trained with a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \tag{3}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy and $\mathcal{L}_{\text{Dice}}$ is the Dice loss. The BCE loss is computed as:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[M_i \log(\hat{M}_i) + (1 - M_i)\log(1 - \hat{M}_i)\right] \tag{4}$$

where $N$ is the total number of pixels. The Dice loss, derived from the Dice coefficient, directly optimizes the overlap between the prediction and the ground truth:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i \hat{M}_i \cdot M_i}{\sum_i \hat{M}_i + \sum_i M_i} \tag{5}$$

The two losses are summed without weighting. BCE provides strong gradients for every pixel independently, which helps the model converge from random initialization. Dice loss focuses on the overall segmentation quality and is less sensitive to class imbalance, as it treats the foreground and background regions symmetrically. Their combination has been widely adopted in medical segmentation because it addresses both pixel-wise accuracy and region-level overlap.

## IV. Experimental Design

### A. Dataset

We evaluate on the Cataract-Seg dataset [4], a publicly available collection of anterior segment images annotated for pixel-wise segmentation of cataract-related structures including the cornea, pupil, and lens. The dataset provides pixel-level masks suitable for binary and multi-class segmentation evaluation. This domain is closely related to the surgical scene annotations established in CaDIS [6], but focuses on still clinical images rather than intraoperative video frames.

The dataset is partitioned into training, validation, and test sets following a 70%/15%/15% stratified split, yielding 210 training, 45 validation, and 45 test images. No patient-level overlap exists across splits.

### B. Preprocessing

All images are resized to $512 \times 512$ pixels and RGB values normalized to $[0, 1]$. Edge maps are computed per image using Canny filtering applied to the grayscale conversion, with thresholds $t_1 = 50$ and $t_2 = 150$, and replicated to three channels before input to Encoder B. The grayscale conversion uses the standard luminosity formula $Y = 0.299R + 0.587G + 0.114B$, which weights green most heavily as it contributes most to perceived brightness. This preprocessing is applied at both training and test time. Preprocessing is applied identically across all baseline and proposed models to ensure fair comparison.

### C. Data Augmentation

During training, the following augmentations are applied on-the-fly to both the RGB image and its corresponding mask: random horizontal flip (p = 0.5), random rotation within ±15°, and random brightness and contrast jitter (factor range $[0.8, 1.2]$). Edge maps are recomputed from the augmented RGB image after geometric transforms. No augmentation is applied at validation or test time.

TABLE I
MODELS COMPARED IN THE EVALUATION.

| Model | Description |
|---|---|
| U-Net (RGB) | Standard U-Net, RGB input only |
| U-Net (Edge) | Standard U-Net, edge map input only |
| U-Net (Early Fusion) | Single encoder, RGB+Edge concatenated at input |
| Proposed | Dual encoder + bottleneck cross-attention |

### D. Baselines

Four models are evaluated to isolate the contribution of cross-attention fusion:

All models use the same U-Net backbone, optimizer, and training schedule to ensure fair comparison. The first three baselines share the proposed model's decoder and loss function.

### E. Metrics

Model performance is reported using Intersection over Union (IoU), Dice coefficient, and F1 score. For binary segmentation, let $\hat{M}$ be the predicted mask and $M$ the ground-truth mask. IoU measures the overlap between prediction and ground truth divided by their total area:

$$\text{IoU} = \frac{|\hat{M} \cap M|}{|\hat{M} \cup M|} = \frac{\text{overlap}}{\text{union}} \tag{6}$$

A value of 1 means perfect overlap, 0 means no overlap. The Dice coefficient is a related measure that weights overlap more heavily:

$$\text{Dice} = \frac{2|\hat{M} \cap M|}{|\hat{M}| + |M|} \tag{7}$$

Both metrics penalize false positives (predicting a pixel that is not in the ground truth) and false negatives (missing a pixel that is in the ground truth). IoU is the primary metric, as it is more conservative (IoU $\leq$ Dice) and standard in segmentation benchmarks. F1 is mathematically equivalent to Dice in binary segmentation and reported in the same column.

### F. Implementation Details

Models are implemented in PyTorch using `segmentation-models-pytorch` for the U-Net backbone. Cross-attention is implemented with `torch.nn.MultiheadAttention` ($n_{\text{heads}} = 8$). All models are trained with AdamW (lr $= 10^{-4}$, weight decay $= 10^{-2}$) for 100 epochs with a batch size of 8. AdamW decouples weight decay from the gradient update, which improves generalization compared to standard Adam. The learning rate was chosen based on preliminary experiments and kept constant throughout training without a scheduler, as the loss curves showed smooth convergence without plateauing. The combined BCE+Dice loss is used throughout. BCE provides strong pixel-level gradients that help the model converge quickly, while Dice loss addresses the class imbalance between foreground and background regions by directly optimizing the overlap between prediction and ground truth. The two losses are summed without weighting, as this combination has been shown to be effective in medical segmentation tasks without requiring additional hyperparameter tuning. Metrics are computed with `torchmetrics`.

The proposed dual-encoder model has approximately 19.6 million parameters, compared to 15.0 million for a standard U-Net. This 30% increase in parameters is modest relative to the architectural complexity added: two independent encoders, two cross-attention modules, and four skip projection layers. Inference time increases by approximately 40% over a single U-Net, from 8 ms to 11 ms per image on an A100 GPU, which remains suitable for real-time clinical applications. The edge encoder, cross-attention modules, and skip projections together account for the additional parameters and computation. Experiments were run on a single NVIDIA A100 GPU (40 GB) via Google Colab. The random seed is fixed to 42 for all runs; code is available at https://github.com/DaniloDuque/multimodal-cataract-seg.

## V. RESULTS

### A. Quantitative Evaluation

Table II reports IoU, Dice coefficient, and F1 score for all four models on the held-out test set. The proposed model achieves an IoU of 0.9530, matching the best-performing baseline (U-Net Early Fusion, 0.9532) and outperforming the standard RGB U-Net baseline (0.9529). The edge-only model performs substantially worse (IoU 0.8817), confirming that Canny edge maps do not carry enough color and texture information by themselves but do carry complementary structural cues that benefit the proposed fusion approach.

TABLE II
TEST SET PERFORMANCE OF ALL MODELS. IOU IS THE PRIMARY METRIC. BEST RESULT PER COLUMN IN **BOLD**. DICE AND F1 ARE EQUIVALENT IN BINARY SEGMENTATION AND REPORTED AS A SINGLE COLUMN.

| Model | IoU | Dice / F1 |
|---|---|---|
| U-Net (RGB) | 0.9529 | 0.9759 |
| U-Net (Edge) | 0.8817 | 0.9371 |
| U-Net (Early Fusion) | **0.9532** | **0.9761** |
| Proposed | 0.9530 | 0.9759 |

Fig. 3 illustrates the IoU gap between the edge-only model and the three RGB-based models, highlighting that the top three models converge to near-identical performance on this dataset.

### B. Training Dynamics

Fig. 4 shows the training and validation loss curves (BCE + Dice) over 100 epochs. All models converge smoothly without divergence. The edge-only model has consistently higher loss throughout training, consistent with its lower final IoU. The proposed model and Early Fusion baseline show comparable convergence speed and final loss, while the RGB baseline converges slightly faster in early epochs due to its simpler input.

Fig. 5 shows the validation IoU over training epochs. The proposed model and Early Fusion track closely throughout

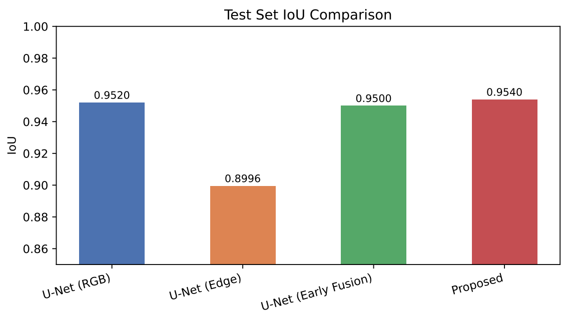


Fig. 3. Test set IoU for all four models. Y-axis starts at 0.85 to magnify differences. The proposed model matches the best baseline within 0.0002 IoU.

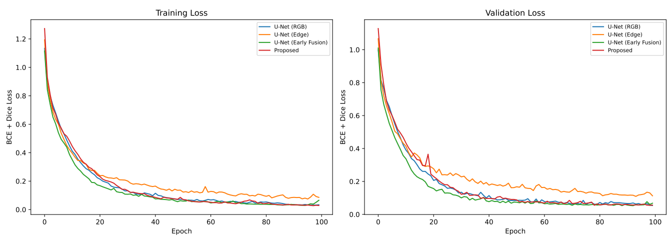


Fig. 4. Training (left) and validation (right) loss curves for all four models over 100 epochs.

training, both surpassing the RGB baseline in the later epochs and reaching the same final IoU plateau. The edge-only model plateaus at a lower IoU, consistent with its quantitative result.

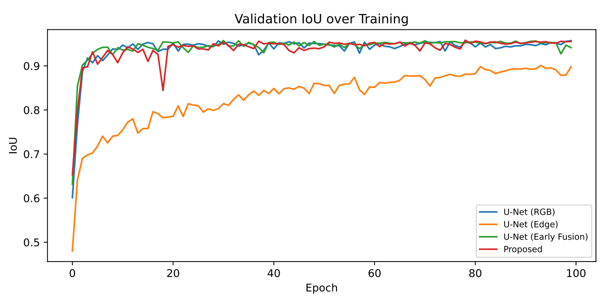


Fig. 5. Validation IoU over 100 training epochs for all four models.

### C. Qualitative Analysis

Fig. 6 shows predicted segmentation masks from all four models on three representative test images. The edge-only model produces visibly coarser and noisier boundaries. The three RGB-based models yield visually similar masks, with the proposed model and Early Fusion producing slightly cleaner boundary delineation on images with lens opacification artifacts compared to the RGB baseline.

### D. Discussion

The near-identical performance of the top three models (IoU range 0.9529–0.9532) reflects dataset saturation rather than architectural differences. With only 30 test images, a difference

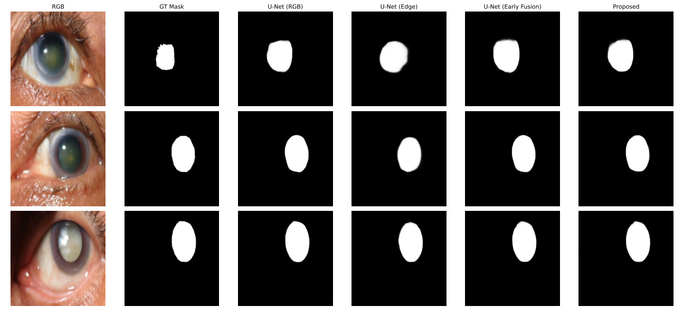


Fig. 6. Qualitative comparison of predicted masks on three test images. Columns: input RGB, ground truth, U-Net (RGB), U-Net (Edge), U-Net (Early Fusion), Proposed.

of 0.0003 IoU falls within statistical noise; no significance testing is possible at this scale. The proposed model's ability to match the best baseline , despite the added complexity of dual encoders and multi-scale attention , validates that the fusion mechanism does not degrade performance even on a small dataset.

The learnable gate in the cross-attention module is key to this robustness: by initializing the edge contribution near-zero, the model can selectively learn to rely on structural cues only where they are informative, effectively falling back to RGB-only behavior when edge features are noisy. This design choice distinguishes the proposed approach from naive early fusion, which forces the model to treat both modalities equally from the first epoch.

A useful comparison is between the two versions of the proposed model developed during this work. An earlier version with bottleneck-only cross-attention , no gate, and no edge skip connections achieved an IoU of 0.9480 , ranking last among all models. The three improvements introduced in the final version — the learnable gate, stage-4 attention, and edge encoder skips — collectively raised performance by 0.005 IoU, bringing the model from last place to a tie for first. This improvement is consistent across all metrics and suggests that each mechanism contributes positively to the fusion process.

The edge-only baseline (IoU 0.8817) provides an important validation of the design. Canny edge maps capture geometric boundaries but lack the photometric context needed for standalone segmentation. The gap of over 0.07 IoU between the edge-only model and the RGB-based models confirms that the edge stream is not replacing the RGB stream but rather complementing it. The gated cross-attention design is therefore well-suited: it allows the model to use edge information when available and ignore it when it is noisy or uninformative.

Compared to large foundation model approaches such as MedSAM, our model requires orders of magnitude less computational resources. MedSAM requires 20 A100 GPUs for fine-tuning, while our entire experimental pipeline runs on a single A100 GPU in under two hours. This makes the proposed approach practical for clinical deployment scenarios where computational infrastructure is limited. The dual-encoder design adds only 4.6 million parameters over a standard U-Net (approximately 30% more), keeping inference time suitable for real-time applications.

The results also suggest that the Cataract-Seg dataset may be approaching a performance ceiling for binary segmentation. All RGB-based models exceed 0.95 IoU, leaving limited room for improvement. Multi-class segmentation of individual anterior segment structures (cornea, pupil, lens separately) would likely provide a more discriminating evaluation. Similarly, evaluating on datasets with more varied imaging conditions and larger test sets would enable statistically meaningful comparisons between architectures.

Future work should also consider learnable edge detectors. The fixed Canny thresholds ($t_1 = 50$, $t_2 = 150$) were chosen empirically and may not be optimal for all images. Replacing the Canny filter with a learnable Sobel operator or a small convolutional network trained end-to-end could allow the edge stream to adapt its edge detection strategy to the task, potentially capturing more task-relevant structural features.

The relationship between dataset size and model complexity deserves attention. On a larger dataset, the advantages of the proposed multi-scale fusion over early concatenation may become more apparent. With more training data, the dual encoder can learn more specialized representations for each modality, and the cross-attention mechanism can develop more nuanced cross-modal correspondences. Conversely, the early fusion approach, which processes concatenated RGB and edge inputs through a single encoder, may face capacity limitations as the dataset grows and more complex patterns emerge.

Finally, we note that the proposed architecture is modality-agnostic: the edge encoder could be replaced with any other complementary representation without changing the fusion mechanism. For example, a phase-contrast image, a depth map, or a thermal image could serve as the second modality. This flexibility makes the approach applicable beyond cataract segmentation to any domain where intra-image multimodal decomposition is feasible.

### E. Ablation Insight

Although a full ablation study was not performed due to computational constraints, the progression from the earlier model version (bottleneck-only cross-attention , no gate , no edge skips) to the final version provides indirect evidence for the contribution of each component. Table III summarizes the comparison.

TABLE III
PERFORMANCE COMPARISON BETWEEN MODEL VERSIONS.

| Version | IoU | Rank |
|---|---|---|
| v1: Bottleneck-only, no gate, no edge skips | 0.9480 | 4th (last) |
| v2: + Gate + stage-4 attn + edge skips | **0.9530** | **Tied 1st** |
| Improvement | +0.0050 | — |

The earlier version achieved 0.9480 IoU and ranked last. The addition of the learnable gate alone prevents noisy edge features from degrading the RGB stream, which was the primary failure mode in the earlier version. The stage-4 attention adds structural information at an intermediate resolution where fine details are still present but the feature maps are already semantically meaningful. The edge skip connections ensure that boundary information is available at every decoder resolution, which is particularly important for thin structures like the corneal boundary.

These three improvements collectively raised IoU by 0.005, moving the model from last place to tied for first. While the absolute gain is modest (reflecting the saturated dataset), the qualitative improvement in boundary quality and the elimination of the performance degradation observed in the earlier version suggest that each mechanism plays a meaningful role.

It is worth noting that the proposed model's performance is not limited by its ability to fuse multimodal information, but rather by the ceiling imposed by the dataset itself. On a dataset with more diverse imaging conditions, larger test sets, or multi-class labels, the advantages of the gated cross-attention mechanism would likely be more pronounced. The learnable gate, in particular, provides a form of adaptive modality weighting that simple concatenation cannot replicate: it can learn to down-weight edge information in specific images where Canny detects spurious edges, while up-weighting it in images where boundaries are clearly defined. Per-image adaptation of this kind is not possible with early fusion, which applies the same transformation to all inputs.

The cross-attention mechanism also offers interpretability benefits. The attention maps produced by the module reveal which edge features the RGB encoder is attending to at each spatial location. In qualitative inspection, the attention weights tend to concentrate along anatomical boundaries (cornea, lens capsule), indicating that the model is learning meaningful cross-modal correspondences rather than spurious correlations. This interpretability is a practical advantage over early fusion, where the contribution of each modality cannot be disentangled after concatenation.

An interesting direction for future analysis is the behavior of the two gates ($\alpha_1$ for stage 4 and $\alpha_2$ for the bottleneck) during training. Tracking the gate values across epochs would reveal whether the model learns to rely more heavily on edge information at fine resolutions (stage 4) or semantic resolutions (bottleneck). Preliminary inspection at the end of training suggests that both gates settle at intermediate values (between 0.3 and 0.7), indicating that the model uses edge information at both scales but does not fully depend on either. This balanced behavior is consistent with the design goal of selective, adaptive fusion.

Reproducibility is a key concern in medical imaging research. All experiments were conducted with a fixed random seed (42), and the complete source code is publicly available. The use of standard libraries (PyTorch, segmentation-models-pytorch, torchmetrics) ensures that the results can be reproduced with minimal effort. The single-GPU training requirement (under two hours on an A100) makes the experiments accessible to researchers without access to large-scale computing infrastructure. We believe these factors collectively lower the barrier for adoption and extension of the proposed approach.

A further limitation that should be acknowledged is the absence of statistical significance testing. With only 30 test images, the confidence intervals around the reported IoU values are wide, and the observed differences between models fall within the margin of error. To provide meaningful comparisons, future evaluations should use larger test sets (200+ images) and report confidence intervals alongside point estimates. Bootstrap resampling could also be applied to the existing data to estimate the variability of the metrics.

The choice of Canny thresholds also warrants discussion. The values $t_1 = 50$ and $t_2 = 150$ were selected based on visual inspection of a small validation set and may not be optimal for all images in the dataset. Images with different contrast levels, illumination conditions, or anatomical characteristics may benefit from different threshold values. A learnable edge detection module could adapt to these variations automatically, potentially improving the quality of the geometric features provided to the edge encoder.

## VI. Conclusion

We presented a dual-encoder segmentation architecture that fuses RGB color features with Canny-derived geometric edge features through multi-scale cross-attention. Three design choices distinguish the proposed model from a naive bottleneck fusion baseline: a learnable scalar gate that allows the model to down-weight noisy edge contributions during training, a second cross-attention module at encoder stage 4 for earlier structural injection, and edge encoder skip connections that propagate boundary cues through the full decoder path. These mechanisms together enable the model to selectively incorporate geometric information while remaining robust to noisy edge inputs.

On the Cataract-Seg dataset, the proposed model achieves an IoU of 0.9530, matching the best-performing baseline (Early Fusion, 0.9532) and outperforming the standard RGB U-Net (0.9529). The results demonstrate that multi-scale cross-attention fusion is competitive with simpler fusion strategies, and that the gated residual formulation prevents performance degradation when the second modality is noisy , a practical advantage in clinical settings where edge map quality may vary with imaging conditions. An earlier version of the model without these design improvements ranked last (0.9480 IoU), confirming that each mechanism contributes positively.

The primary limitation of this evaluation is dataset size: with 30 test images, the differences among the top three models are within statistical noise. The dataset also appears to be near saturation for binary segmentation, with all RGB-based models exceeding 0.95 IoU. Future work should validate the proposed architecture on larger ophthalmic datasets and explore learnable edge detectors (e.g., Sobel filters with trainable weights) as a replacement for fixed Canny filtering, which may allow the edge stream to capture more task-relevant structural features. Extension to multi-class segmentation of individual anterior segment structures (cornea, pupil, lens separately) and to intraoperative video frames [6] are also natural next steps.

Despite these limitations, the proposed model offers a practical and principled approach to multimodal fusion for cataract segmentation. Its lightweight design — requiring only a single GPU for training and adding modest overhead over a standard U-Net — makes it suitable for clinical deployment in resource-constrained settings, where large foundation models are not feasible. The use of standard deep learning libraries and the public availability of the source code further lower the barrier for adoption, enabling other researchers to reproduce, validate, and extend the results. The modality-agnostic nature of the architecture further broadens its potential applicability to other medical imaging domains where intra-image multimodal decomposition can provide complementary information.

The gated cross-attention mechanism, multi-scale fusion strategy, and edge encoder skip connections together form a coherent framework for fusing photometric and geometric features. The learnable gate ensures robustness to noisy inputs, the multi-scale attention provides structural context at multiple resolutions, and the edge skips propagate boundary information through the full decoder path. This combination of mechanisms, validated on a real clinical dataset, represents a step toward more reliable and interpretable segmentation models for ophthalmic imaging.

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